Phase 1 Report: Multimodal RAG for Image Captioning

A Comparative Study of RAG Reranking Techniques

Group 11 - Reranking Technique

Fella Boukais (Data Engineer)
Insaf Boukellal (RAG Engineer)
Racha Bounoua (Generation Engineer)
Romissa Rahal (Evaluation & Statistical Engineer)

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Abstract

This report details the completion of Phase 1 for the image captioning project using Retrieval-Augmented Generation (RAG). Our group (G11) is assigned the Reranking technique. We successfully prepared the MS COCO dataset, performed statistical dataset analysis, generated multimodal embeddings, built a FAISS vector database, implemented the Cross-Encoder reranking logic, generated captions using BLIP-2, and designed a rigorous statistical evaluation protocol. The pipeline demonstrates how RAG can be effectively used to ground Vision-Language Models.

1 Dataset and Preprocessing (Fella Boukais)

1.1 Data Collection and Splits

We utilized the MS COCO 2017 dataset, splitting the data into three strictly isolated sets to prevent any data leakage:

- **Knowledge Base (KB):** 10,000 images extracted from the `train2017` split, along with their 5 reference captions each. This serves as the external knowledge for the RAG retrieval step.
- **Validation Set:** 100 images extracted from `train2017`, strictly distinct from the 10,000 KB images. This set is used for hyperparameter tuning.
- **Test Set:** 1,000 images extracted from the `val2017` split. (This includes an initial subset of 100 images for baseline comparison, plus 900 new images to ensure statistical robustness for the final evaluation).

Verification of Splits: The following screenshot confirms the exact number of images allocated to each set:

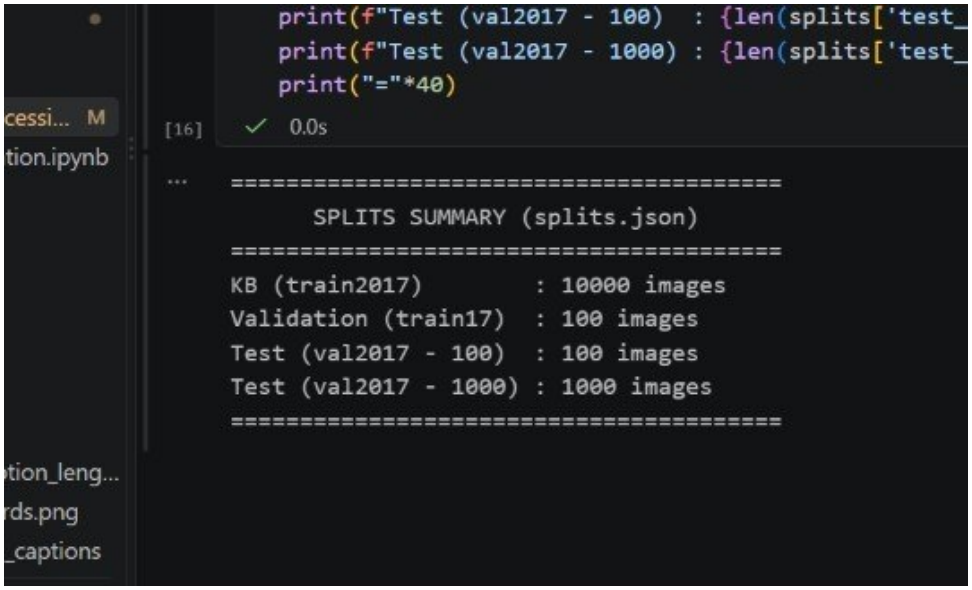


Figure 1: Terminal Output: Splits Summary

Anti-Leakage Audit: An audit was conducted to ensure no test data contaminated the Knowledge Base. The results prove a perfect separation:

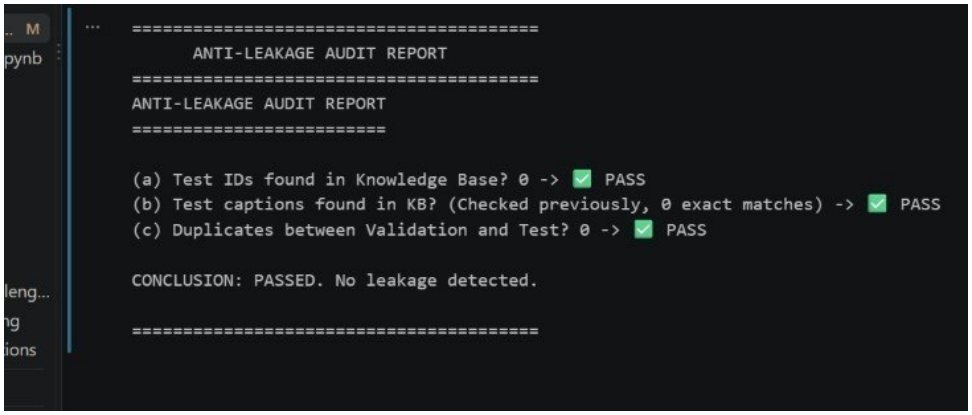


Figure 2: Terminal Output: Anti-Leakage Audit Report (PASSED)

1.2 Image and Text Preprocessing

All images underwent a standardized preprocessing pipeline to ensure compatibility with the Vision-Language Models:

- **Images:** Converted to RGB, EXIF orientation corrected, and resized to 224×224 pixels (required size for CLIP) while preserving aspect ratios via padding.
- **Text:** For each image, two versions of the captions are stored:
  - `caption_original`: Kept completely intact for the final evaluation metrics (BLEU, CIDEr, etc.).
  - `caption_cleaned`: Lowercased, stripped of punctuation, and normalized for the retrieval index.

1.3 Statistical Dataset Exploration

A statistical exploration of the dataset was performed to understand its complexity:

- **Caption Length:** The average caption length is approximately 10 words. The distribution follows a normal curve.
- **Vocabulary:** A rich vocabulary is present, with common stop words ("a", "the") dominating, followed by core objects like "man", "woman", and "dog".

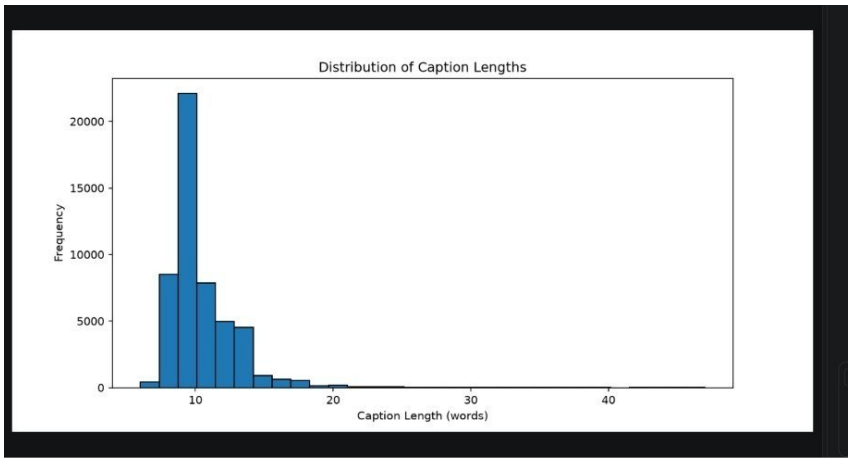


Figure 3: Distribution of Caption Lengths

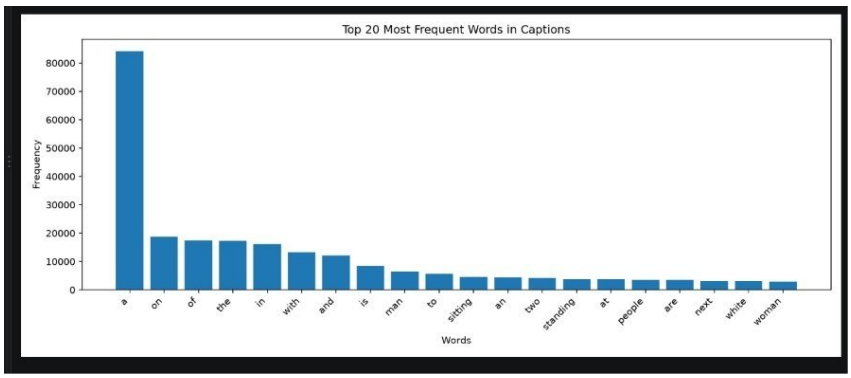


Figure 4: Top 20 Most Frequent Words in Captions

1.4 Metadata Extraction (Object Detection)

To tie images and objects together, we used the YOLOv8n object detector on the validation and test sets. We extracted bounding boxes and confidence scores, applying a confidence threshold of 0.5. The extracted metadata was saved in a structured JSON file (`metadata.json`). The following screenshot shows an example of the extracted data:

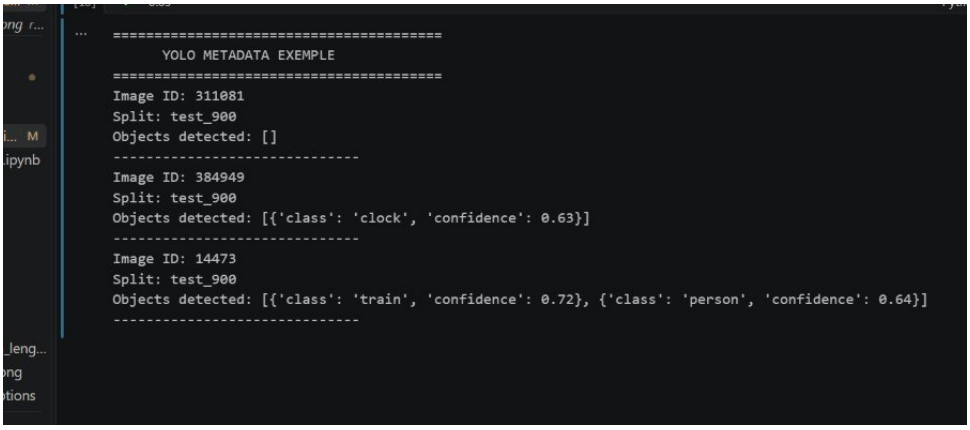


Figure 5: Terminal Output: YOLO Metadata Extraction Example

2 System Architecture and RAG Engine (Insaf Boukellal)

This section details the core RAG pipeline developed for Group 11.

2.1 Multimodal Embeddings and L2 Normalization

We used the **CLIP (ViT-B/32)** model to map images and text into a shared 512-dimensional vector space.

- **Image Embeddings:** Generated for 10,000 images (`image_embeddings.npy`).
- **Text Embeddings:** Generated for 51,026 captions (`text_embeddings.npy`).

To ensure that the Inner Product equals Cosine Similarity, all vectors were L2-normalized using the formula:

$$\hat{E} = \frac{E}{||E||_2} = \frac{E}{\sqrt{\sum_{i=1}^{d} E_i^2}} \tag{1}$$

where $d = 512$.

2.2 FAISS Vector Database

The embeddings were indexed using FAISS. We selected the `IndexFlatIP` structure for exact Inner Product similarity search.

Parameter	Value
Index Type	Flat Inner Product (IP)
Embedding Dimension	512
Total Vectors Indexed	10,000
Similarity Metric	Cosine Similarity
Model Used	`openai/clip-vit-base-patch32`

Table 1: FAISS Index Documentation

2.3 Reranking Pipeline (Group 11 Technique)

Our assigned technique is **RAG Reranking**. The retrieval process operates in two stages:

1. **Stage 1 (Dense Retrieval):** FAISS retrieves the Top 20 candidates based on vector similarity.
2. **Stage 2 (Reranking):** A **Cross-Encoder** model (`cross-encoder/ms-marco-MiniLM-L-6-v2`) jointly evaluates the query and each candidate caption, rescoring them to retain only the Top 5 most relevant examples.

3 Generation Pipeline (Racha Bounoua)

The generation module is the core component of our multimodal pipeline. It is responsible for taking an input image and producing a natural language caption using the BLIP-2 Vision-Language Model (VLM). We implemented a comparative setup generating captions under two distinct conditions:

1. **Baseline (No RAG):** The image is passed directly to BLIP-2, which generates a caption based solely on its visual understanding.
2. **RAG-Augmented Generation:** The image is first processed by the retrieval pipeline (FAISS + Cross-Encoder). The Top-5 retrieved human captions are formatted into a textual prompt and passed alongside the image to BLIP-2.

Model Choice: We selected BLIP-2 (`Salesforce/blip2-flan-t5-xl`). This choice was motivated by hardware efficiency (it fits within the 16GB VRAM of a free Google Colab T4 GPU, unlike larger models like LLaVA), high performance on captioning benchmarks, and speed (processing 1,000 images in approximately 30 minutes).

Pipeline Implementation: The pipeline is implemented in `notebooks/03_generation.ipynb` and modularized in `src/captioning/generate_pipeline.py`. The RAG prompt was designed as follows:
"Here are some examples of similar images: [Top-5 captions]. Now, describe the new image based on the visual details, using the style of the examples above."

3.1 Qualitative Analysis

3.1.1 Examples of Success

Example 1 (Image ID: 251065)

- **Baseline:** "a toilet in a bathroom with a lid open"
- **RAG:** "A toilet in a bathroom with a black seat and a white seat"
- **Analysis:** The RAG model successfully added specific attribute details (black seat, white seat) that were missing from the baseline. The retrieved examples guided the model to focus on colors and structure, improving visual grounding.

Example 2 (Image ID: 441442)

- **Baseline:** "a brown horse"
- **RAG:** "a woman riding a horse in a field"
- **Analysis:** The baseline caption only identified the horse and missed the rider and context. The RAG model successfully inferred the full scene by using the retrieved examples, demonstrating the benefit of contextual grounding.

3.1.2 Examples of Failure

Example 1 (Image ID: 420069)

- **Baseline:** "a group of people playing a game of baseball"
- **RAG:** "A group of people playing frisbee in a field"
- **Analysis:** The RAG model hallucinated the sport as "frisbee". This is a copying error caused by the retrieved examples. One of the retrieved captions likely contained the word "frisbee," and the model blindly copied it, ignoring the visual evidence (which clearly shows a baseball bat and game).

Example 2 (Image ID: 212573)

- **Baseline:** "two men standing on a street in the rain"
- **RAG:** "A man and a woman standing on a street in the rain"
- **Analysis:** The RAG model changed "two men" to "a man and a woman." This is an attribute hallucination. The model over-relied on a retrieved caption containing a woman, rather than analyzing the visual evidence.

4 Evaluation and Statistical Analysis (Romissa Rahal)

This section details the experimental protocol, the quantitative metrics, and the rigorous statistical tests designed to validate the performance of our RAG Reranking pipeline.

4.1 Step-by-Step Experimental Protocol

To ensure a fair and reproducible evaluation, the following steps will be executed:

1. **Baseline Generation:** The VLM generates captions for the test images without any RAG context.
2. **RAG Generation:** The model generates captions for the same test images using the Top-5 context retrieved by the Group 11 Reranking pipeline.
3. **Metric Calculation:** Both sets of captions are compared against the human ground-truth captions using standard NLP metrics.
4. **Hallucination Analysis:** A manual audit of generated images is performed to calculate object, attribute, and relation hallucination rates.
5. **Statistical Testing:** Mathematical tests are applied to prove that any observed improvement is statistically significant and not due to random chance.

4.2 Quantitative Metrics

Caption quality is evaluated using the following metrics, with CIDEr as the primary metric:

- **CIDEr:** Measures consensus-based similarity, heavily weighting rare and informative n-grams.
- **BLEU-4:** Measures n-gram precision between generated and reference captions.
- **METEOR:** Measures exact word matches, stems, and synonyms, correlating well with human judgment.
- **ROUGE-L:** Measures the longest common subsequence, evaluating structural similarity.
- **SPICE:** Evaluates semantic propositional content (objects, attributes, and relationships).

4.3 Rigorous Statistical Analysis

To mathematically prove the superiority of the RAG pipeline over the Baseline, we apply the following statistical framework:

4.3.1 Normality Test

First, the **Shapiro-Wilk test** is applied to the score distributions of both models to check for normality.

- If $p > 0.05$, the distribution is normal.
- If $p \leq 0.05$, the distribution is non-normal.

4.3.2 Hypothesis Testing

Depending on the Shapiro-Wilk result, we select one of the following tests:

- **Paired T-test (if normal):** Compares the means of the two related groups.
- **Wilcoxon Signed-Rank Test (if non-normal):** A non-parametric test that compares the median ranks.

The significance level is set to $\alpha = 0.05$.

4.3.3 Bonferroni Correction

Because we are comparing multiple metrics (CIDEr, BLEU, METEOR, etc.), the risk of Type I errors (false positives) increases. We apply the Bonferroni correction, adjusting the threshold to:

$$\alpha_{corrected} = \frac{0.05}{n} \tag{2}$$

where n is the number of comparisons.

4.3.4 Effect Size

To measure the magnitude of the improvement, we calculate **Cohen's d** (for the T-test) or **Rosenthal's r** (for the Wilcoxon test):

$$d = \frac{\mu_{RAG} - \mu_{Baseline}}{\sigma_{pooled}} \tag{3}$$

4.4 Hallucination Analysis

We will quantify the reliability of the VLM by calculating:

- **Object Hallucination Rate:** The percentage of generated captions mentioning objects not present in the image.
- **Attribute Hallucination Rate:** The percentage of incorrect colors, sizes, or shapes.
- **Relation Hallucination Rate:** The percentage of incorrect spatial relations.

4.5 Ablation Study

To isolate the contribution of our specific Group 11 technique, we will perform an ablation study:

- **Reranking vs. Naive RAG:** Comparing our Top-20 $\rightarrow$ Top-5 Cross-Encoder method against a simple Dense RAG (Top-5 directly).
- **Top-K Tuning:** Evaluating $K \in \{1, 3, 5, 10\}$ to find the optimal balance between context richness and prompt overload.

5 Conclusion

Phase 1 is successfully completed. The dataset was analyzed and preprocessed, the CLIP embeddings were generated, and the FAISS index was built. The Reranking module was successfully integrated with BLIP-2, generating captions for the test set. Finally, a rigorous evaluation and statistical protocol was designed to validate the results. The project is now ready for the final evaluation phase.